

Group Assignment I

Advanced Econometrics 2026–2027

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Question 1

First write out $\sigma_{t+1}^2(\sigma_t^2, \epsilon_t)$, i.e. write the volatility equation as a function of the innovations.

We know:

$$x_t = \mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t$$

and by the given equation for σ_t we know:

$$\sigma_{t+1}^2 = \omega + \left(\alpha + \delta \tanh(-\gamma x_t) \right) \left(\frac{x_t - \mu - \lambda\sigma_t^2}{\sigma_t} \right)^2 + \beta\sigma_t^2.$$

Substituting now gives us:

$$\begin{aligned} \sigma_{t+1}^2 &= \omega + \left(\alpha + \delta \tanh \left(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) \right) \right) \left(\frac{(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) - \mu - \lambda\sigma_t^2}{\sigma_t} \right)^2 + \beta\sigma_t^2 \\ &= \omega + \left(\alpha + \delta \tanh \left(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) \right) \right) \left(\frac{\sigma_t\epsilon_t}{\sigma_t} \right)^2 + \beta\sigma_t^2 \\ &= \omega + \left(\alpha + \delta \tanh \left(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) \right) \right) \epsilon_t^2 + \beta\sigma_t^2. \end{aligned}$$

We have now written $\sigma_{t+1}^2(\sigma_t, \epsilon_t) = \omega + \left[\alpha + \delta \tanh \left(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) \right) \right] \epsilon_t^2 + \beta\sigma_t^2$. We want to show that $\{\sigma_t\}_{t \in \mathbb{Z}}$ is SE. We can do this by Bougerol's Theorem.

A1 $\{\epsilon_t\}_{t \in \mathbb{Z}}$ is SE. This holds since $\{\epsilon_t\} \sim TID(v)$ is iid and any iid sequence is SE.

A2 There should exist a finite initialization σ_1^2 such that $\mathbb{E}[\log^+ |\sigma_2^2(\sigma_1^2, \epsilon_1)|] < \infty$. We have:

$$\begin{aligned} \mathbb{E}[\log^+ |\sigma_2^2(\sigma_1^2, \epsilon_1)|] &= \mathbb{E}\left[\log^+ \left| \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1))] \epsilon_1^2 + \beta\sigma_1^2 \right| \right] \\ &\leq \mathbb{E}\left[\left| \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1))] \epsilon_1^2 + \beta\sigma_1^2 \right| \right] \quad (\log^+ |z| \leq |z|) \\ &\leq |\omega| + \mathbb{E}\left[\left| \alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1)) \right| \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (\text{triangle inequality}) \\ &\leq |\omega| + \mathbb{E}\left[(|\alpha| + |\delta| |\tanh(\dots)|) \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (\text{triangle inequality}) \\ &\leq |\omega| + \mathbb{E}[(|\alpha| + |\delta|) \epsilon_1^2] + |\beta| |\sigma_1^2| \quad (-1 \leq \tanh(\cdot) \leq 1) \\ &= |\omega| + (|\alpha| + |\delta|) \mathbb{E}[\epsilon_1^2] + |\beta| |\sigma_1^2| \\ &< \infty. \end{aligned}$$

This is finite since $\omega, \alpha, \beta, \delta$ are finite parameters, $\mathbb{E}[\epsilon_1^2] < \infty$ for $\epsilon_1 \sim TID(v)$ with $v > 2$, and σ_1^2 is a finite initialization. So A2 holds.

A3 The contraction holds when

$$\mathbb{E} \left[\log \sup_{x \in \mathcal{X}} \left| \frac{\partial \phi(x, \epsilon_t)}{\partial x} \right| \right] < 0.$$

Here $\phi(\sigma_t^2, \epsilon_t) = \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t))] \epsilon_t^2 + \beta\sigma_t^2$.

We recall also that $\frac{d \tanh(x)}{dx} = 1 - \tanh^2(x)$.

By the chain rule,

$$\frac{\partial \phi(\sigma_t^2, \varepsilon_t)}{\partial \sigma_t^2} = \delta \varepsilon_t^2 \left(1 - \tanh^2(-\gamma(\mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t)) \right) \cdot (-\gamma) \left(\lambda + \frac{\varepsilon_t}{2\sqrt{\sigma_t^2}} \right) + \beta.$$

We then bound,

$$\begin{aligned} \left| \frac{\partial \phi(\sigma_t^2, \varepsilon_t)}{\partial \sigma_t^2} \right| &\leq |\gamma| |\delta| \varepsilon_t^2 \left| 1 - \tanh^2(-\gamma(\mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t)) \right| \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\sigma_t^2}} \right) + |\beta| \quad (\text{triangle inequality}) \\ &\leq |\gamma| |\delta| \varepsilon_t^2 \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\sigma_t^2}} \right) + |\beta| \quad (0 \leq 1 - \tanh^2(\cdot) \leq 1) \\ &\leq |\gamma| |\delta| \varepsilon_t^2 \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\omega}} \right) + |\beta| \quad (\sigma_t^2 \geq \omega \Rightarrow \frac{1}{2\sqrt{\sigma_t^2}} \leq \frac{1}{2\sqrt{\omega}}) \\ &= |\gamma| \left(|\lambda \delta| \varepsilon_t^2 + \frac{|\delta| |\varepsilon_t|^3}{2\sqrt{\omega}} \right) + |\beta|, \end{aligned}$$

where $\sigma_t^2 \geq \omega$ holds for all t since $\alpha > |\delta|$ and $|\tanh(\cdot)| \leq 1$, $\alpha + \delta \tanh(-\gamma x_{t-1}) \geq \alpha - |\delta| > 0$, so the middle term of the σ_t^2 -recursion is non-negative and since $\beta \geq 0$ so is $\beta \sigma_{t-1}^2$. Hence $\sigma_t^2 \geq \omega$.

Therefore,

$$\mathbb{E} \left[\log \sup_{x \in \mathcal{X}} \left| \frac{\partial \phi(x, \varepsilon_t)}{\partial x} \right| \right] \leq \mathbb{E} \left[\log \left(|\gamma| \left(|\lambda \delta| \varepsilon_t^2 + \frac{|\delta| |\varepsilon_t|^3}{2\sqrt{\omega}} \right) + |\beta| \right) \right] < 0,$$

which is exactly the condition given in the question.

Thus, by Bougerol's theorem, $\{\sigma_t^2\}_{t \in \mathbb{Z}}$ converges to a SE process. Since $x_t = \mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t$ is a measurable function of the jointly SE sequence $\{(\sigma_t, \varepsilon_t)\}_{t \in \mathbb{Z}}$, in this case $\{x_t\}_{t \in \mathbb{Z}}$ is also SE by Krengel's theorem.

Question 2

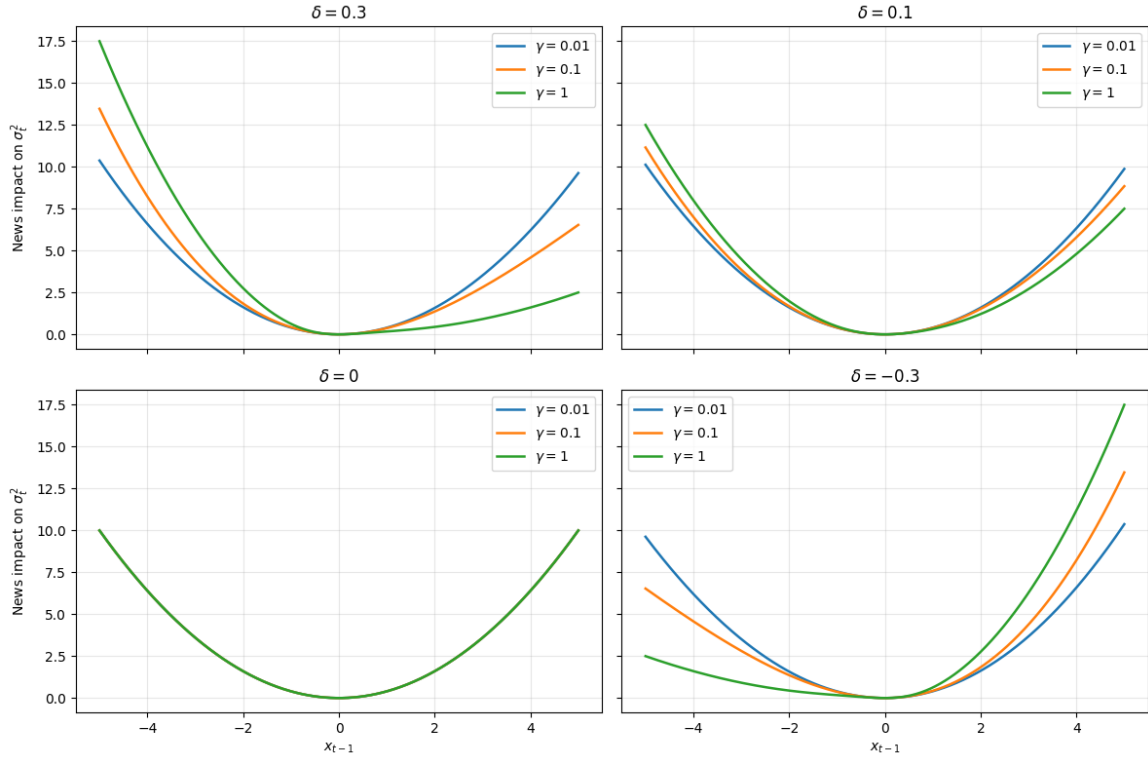


Figure 1: News impact curves of the GARCH-M-L model for different values of δ and γ .

The news impact expression is given by $\text{NIC}(x_{t-1}) = (\alpha + \delta \tanh(-\gamma x_{t-1}))x_{t-1}^2$. Looking at Figure 1, it can be seen that different values of δ and γ have a significant impact on the shape of the news impact curve. When $\delta > 0$, the news impact curve is asymmetric, indicating that negative shocks have a larger effect on volatility than positive shocks of equal size. And $\delta < 0$ mirrors it. When $\delta \rightarrow 0$, all curves collapse to a symmetric one regardless of γ . Indeed, as $\delta = 0$, γ is dropped, the expression reduces to αx_{t-1}^2 , which is a parabola.

Regarding γ , it controls how abruptly the asymmetry switches on: $\gamma = 0.01$ is almost symmetric since $\tanh(x) \approx x$ for relatively small x . When γ increases, the asymmetry becomes more pronounced. As $\gamma \rightarrow \infty$, the $\tanh$ function approaches a step function, giving $\alpha + \delta$ for negative shocks and $\alpha - \delta$ for positive shocks.

Question 3

Table 1: Descriptive statistics of the daily holding period returns

	PFE	JNJ	MRK	AAPL
Number of observations	3270	3270	3270	3270
Mean	0.041	0.046	0.057	0.107
Median	0.000	0.036	0.046	0.089
Standard deviation	1.368	1.076	1.306	1.783
Minimum	-7.735	-10.038	-9.863	-12.865
Maximum	10.855	7.998	10.408	11.981
Skewness	0.223	-0.150	0.083	-0.076
Excess kurtosis	5.294	9.370	6.637	5.360

Table 1 provides descriptive statistics of the daily holding period returns for Pfizer (PFE), Johnson & Johnson (JNJ), Merck & Co. (MRK), and Apple (AAPL). The number of observations is 3270 for each stock, and the mean daily return ranges from 0.041% for PFE to 0.107% for AAPL. In terms of volatility, AAPL has the highest standard deviation while JNJ has the lowest indicating that AAPL is more volatile compared to the other stocks (e.g. JNJ), which can be observed in Figure 2. The minimum and maximum daily returns also vary across the stocks, with AAPL having the largest range of returns, from -12.865% to 11.981%, while JNJ has the smallest range, from -10.038% to 7.998%. The skewness and excess kurtosis values indicate that the return distributions are not perfectly normal, with some stocks exhibiting left-skewed distributions (JNJ and AAPL) and others right-skewed (PFE and MRK). All stocks show excess kurtosis greater than 3, suggesting that the return distributions have heavier tails than a normal distribution.

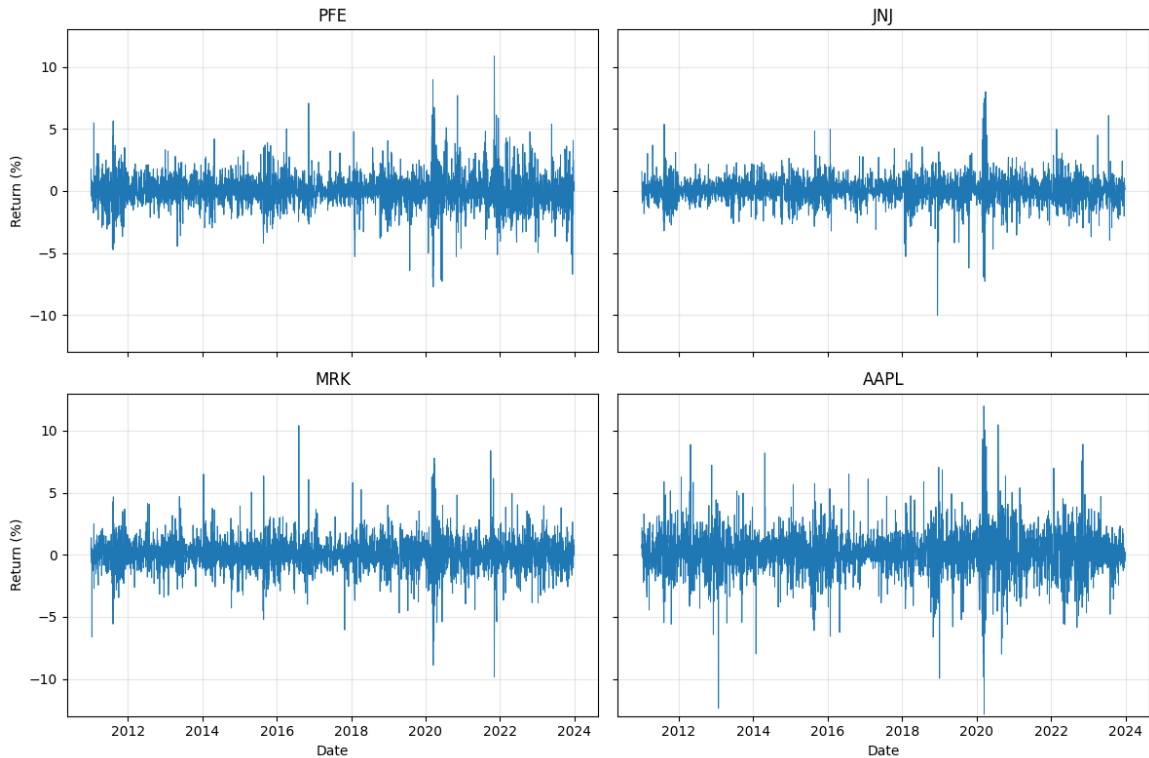


Figure 2: Daily holding period returns of Pfizer, Johnson & Johnson, Merck & Co., and Apple.

It is also remarkable that the returns of all four stocks exhibit volatility clustering, where periods of high volatility are followed by periods of low volatility and vice versa. Looking at the time around the COVID-19 pandemic, we can see that there is a significant fluctuation in volatility for all four stocks, which is consistent with the market-wide uncertainty and turbulence during that period.

Question 4

Table 2: Maximum likelihood estimates of the three volatility models

	PFE			JNJ			MRK		
	M1	M2	M3	M1	M2	M3	M1	M2	M3
μ	0.037	-0.028	-0.031	0.072	0.058	0.033	0.066	0.077	-0.046
λ		0.111	0.104		0.031	0.063		-0.015	0.117
ω	-0.027	-0.028	-0.025	-0.002	-0.002	0.003	0.014	0.015	0.019
α	0.036	0.036	0.043	0.025	0.025	0.026	0.039	0.040	0.040
β	0.961	0.963	0.953	0.923	0.924	0.916	0.906	0.904	0.906
δ			0.029			0.017			0.039
γ			0.410			1.228			4.742
ν	4.651	4.677	4.903	4.394	4.389	4.556	4.498	4.501	4.715
Log-lik	-3769	-3767	-3753	-3283	-3283	-3270	-3893	-3893	-3868
AIC	7549	7546	7522	6576	6577	6557	7796	7798	7752
BIC	7578	7581	7569	6605	6612	6603	7825	7833	7799

Notes: M1, M2, and M3 are denoted as the GARCH, GARCH-M, and GARCH-M-L models, respectively. AIC and BIC are the Akaike information criterion and the Bayesian information criterion, respectively. They are a measure of the model's fit to the data; the lower the value, the better the model.

Table 2 shows the maximum likelihood estimates of the three volatility models for PFE, JNJ, and MRK. Based on the AIC and BIC criteria, we can see that the GARCH-M-L model appears to be the best model for all three stocks, given its lowest AIC and BIC values, followed by the GARCH-M model, and then the GARCH model. Moreover, the log-likelihood values also strengthen this conclusion in which the GARCH-M-L models have the highest log-likelihood values. So, it is clear that including the mean and leverage effects improves the model's fit to the data.

Within the GARCH-M-L model, we can see that the δ parameter is positive for all three stocks, indicating that negative shocks have a larger impact on volatility than positive shocks of equal size. $\hat{\gamma} \approx 4.7$ for MRK, which is much larger than for others, so one can expect that MRK's news impact curve will look almost kinked at zero, while the other two will be more smooth.

Question 5

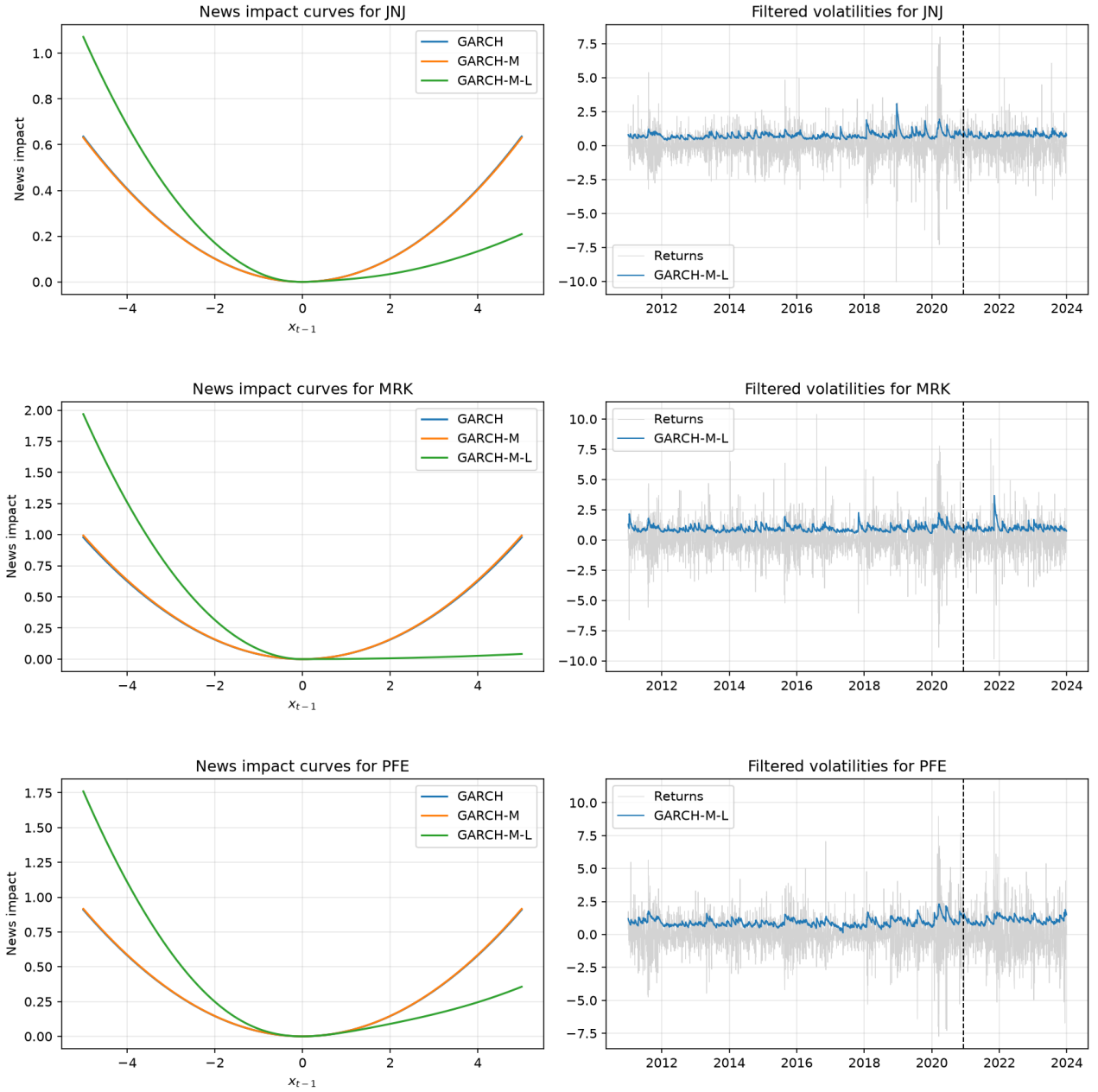


Figure 3: News-impact curves and filtered volatilities. The left panels plot news-impact curve $\text{NIC}(x_{t-1}) = (\alpha + \delta \tanh(-\gamma x_{t-1}))x_{t-1}^2$ for GARCH, GARCH-M, and GARCH-M-L, using each model's estimates from Q4. The right panels plot the observed return x_t (grey) and GARCH-M-L filtered volatility $\hat{\sigma}_t$ (blue) over the full sample; the dashed line marks January 4, 2021, the end of the in-sample window. Stocks: JNJ, MRK, PFE from CRSP daily data, 2011–2023.

The GARCH-M-L news-impact curves are clearly asymmetric for all three stocks, confirming a leverage effect: negative shocks raise conditional volatility more than positive shocks of equal size, most strongly for MRK and PFE. GARCH and GARCH-M curves coincide exactly, since λ does not enter the news-impact function. Consistent with this asymmetry, the filtered volatility for all three stocks rises sharply following large negative return shocks, most visibly around the COVID-19 onset in early 2020, while comparable positive shocks produce smaller increases in filtered volatility. JNJ additionally shows a stock-specific spike in 2019. The filtered volatility after 2021 remains within a similar range to the in-sample volatility, suggesting the estimated dynamics generalize reasonably well out-of-sample.

Question 6

Table 3: Simulation-based Value-at-Risk estimates

Stock	Model	1 step			5 steps			20 steps		
		1%	5%	10%	1%	5%	10%	1%	5%	10%
PFE	M1	-3.78	-2.24	-1.62	-7.93	-4.95	-3.73	-14.47	-9.31	-6.94
PFE	M2	-3.76	-2.19	-1.58	-7.55	-4.74	-3.51	-13.02	-8.12	-5.86
PFE	M3	-3.82	-2.28	-1.63	-7.80	-4.75	-3.45	-13.70	-8.20	-5.80
JNJ	M1	-2.41	-1.42	-1.02	-5.10	-3.13	-2.23	-10.10	-5.57	-3.87
JNJ	M2	-2.50	-1.46	-1.05	-5.29	-3.10	-2.23	-9.29	-5.60	-3.89
JNJ	M3	-2.41	-1.41	-1.00	-5.18	-3.00	-2.16	-9.77	-5.83	-3.91
MRK	M1	-3.00	-1.71	-1.22	-5.92	-3.74	-2.72	-12.02	-7.15	-5.08
MRK	M2	-2.99	-1.71	-1.22	-6.36	-3.82	-2.75	-12.05	-7.39	-5.29
MRK	M3	-3.09	-1.74	-1.27	-6.70	-4.03	-2.90	-12.18	-7.53	-5.45

Notes: The table reports estimated lower-tail Value-at-Risk (VaR) quantiles for the 1-, 5-, and 20-day compound holding-period returns at the 1%, 5%, and 10% probability levels. M1 denotes the GARCH model, M2 the GARCH-M model, and M3 the GARCH-M-L model. Model parameters are estimated using the first 2,500 observations and are kept fixed while the conditional volatility is filtered up to and including January 4, 2021. VaR estimates are based on 10,000 Monte Carlo simulations and are reported in percentage points.